Logo Data Engine 使用說明與資料流規格（LaTeX 風格版）

版本：v1.1（對應 /raid/ming/logo/logo_data_engine）

1. 系統概述

本系統為一個可重跑、可擴充、可審核的 Logo 資料引擎。核心原則為： 所有資料都進入同一 schema（ImageRecord → LogoInstance → BrandRecord）， 並以 `tier` 作為唯一狀態機。

Pipeline 預設包含：Collector、Ontology、Ingest、Quality Gate、Segmentation（SAM3）、Annotation（OCR/CLIP/Caption/VLM）、Review、Export。

2. 環境與啟動

2.1 安裝與更新

```
cd /raid/ming/logo
./logo_data_engine/setup_env.sh logo_sam3
```

此腳本會:

- 建立 conda 環境（Python 3.12）
- 安裝依賴、Torch、PaddleOCR、Ultralytics、HuggingFace 模型
- 下載或更新 SAM3 至 `/raid/ming/logo/logo_data_engine/sam3`

2.2 快速執行（多品牌 batch）

```
./logo_data_engine/run_multibrand_batch.sh \
  --run-name batch_10 \
  --target-records 10 \
  --with-sam3 \
  --use-vlm \
  --vlm-model-id llava-hf/llava-1.5-7b-hf
```

2.3 GPU 指定

```
./logo_data_engine/run_multibrand_batch.sh \
  --run-name batch_10 \
  --target-records 10 \
  --with-sam3 \
  --cuda-devices 1,2,3
```

3. Pipeline 預設流程與觸發

每次執行會先跑 `preflight`，輸出：

```
results/<run_name>/meta/preflight.json
```

內容包含所有模型的可用性與錯誤訊息。

4. 資料輸出流（非常詳實）

4.1 流程順序與輸入/輸出

步驟	輸入	輸出	說明
Collector	品牌/類別/來源設定	meta/raw_records.json fetch/images/	取得原始樣本與下載圖像
Balance (candidate)	raw_records.json	candidate_records.json	候選池平衡抽樣
Ontology Fetch	candidate_records.json	brand/brand_records.json	抓取品牌知識與 alias
Seed Ontology	brand_records.json	DB.brand_records	寫入 brand tier= silver
Ingest	candidate_records.json	DB.image_records	寫入影像表 + pHash 去重
Quality Gate	DB.image_records	meta/passed_records.json quality_status	五關過濾 + 狀態標記

步驟	輸入	輸出	說明
Balance (final)	passed_records.json	fetch/records.json	達成目標筆數
SAM3 Segmentation	fetch/records.json brand_records.json	segment/records_with_logo_masks.json segment/masks segment/visualizations	object-first + SAM3 mask
Phase1 Annotate	fetch/records.json brand_records.json segment records (optional)	DB.logo_instances review/review_queue.json	OCR/CLIP/Caption/VLM merge
Export	DB tables	export/knowledge_base.json analysis/analysis.md analysis/summary.json	KnowledgeBase + 分析報告

5. Quality Gate 詳細評分邏輯

Gate	條件 / 計算	閾值	失敗原因
Gate 1	尺寸檢查：min_width, min_height	≥ 256	too_small
Gate 2	模糊度：Laplacian variance	≥ 45.0	too_blurry
Gate 3	pHash distance	≤ 6	duplicate
Gate 4	CLIP Logo score	hard ≥ 0.22 soft ≥ 0.08	low_clip_logo_score
Gate 5	YOLO Prescreen labels: logo/brand logo/company logo	conf ≥ 0.30	no_logo_prescreen_detection

6. 模型 Prompts 與輸出樣例（詳細）

6.1 GroundingDINO（Logo bbox）

Prompt 來源（由 `build_prompts` 組合）：

- 品牌詞：`{brand} logo`, `{brand} brand logo`, `{brand} emblem`
- Alias：`{alias} logo`
- 物件結合：`{brand} logo on {object}`, `{object} logo`
- 歷史語境：`vintage {brand} logo`, `retro {brand} logo`

範例 Prompt 列表：

```
nike logo
nike brand logo
nike emblem
shoe logo
nike logo on shoe
vintage nike logo
```

輸出樣例（節錄）：

```
{
  "grounding_bbox_xyxy": [112.3, 76.1, 188.4, 132.7],
  "grounding_score": 0.71,
  "grounding_label": "nike logo"
}
```

6.2 GroundingDINO（Object-first 物件框）

Prompt 來源（依 category 與 object_terms）：

- shoes：`shoe`, `sneaker`, `footwear`
- apparel：`shirt`, `hoodie`, `apparel`
- smartphones：`phone back`, `mobile phone`
- 通用：`product`, `object`, `item`

輸出樣例：

```
{
  "object_grounding_bbox_xyxy": [18.5, 44.2, 502.9, 363.7],
  "object_grounding_label": "sneaker",
  "object_grounding_score": 0.83
}
```

6.3 SAM3 Segmentation

輸出欄位：

- `logo_bbox_xyxy`, `logo_mask_path`
- `logo_mask_area_pixels`, `logo_mask_area_fraction`
- `logo_segmentation_status`

輸出樣例：

```
{
  "logo_bbox_xyxy": [120.0, 78.2, 182.4, 124.9],
  "logo_mask_path": ".../segment/masks/record_001.png",
  "logo_mask_area_pixels": 2412,
  "logo_mask_area_fraction": 0.0032
}
```

6.4 PaddleOCR

輸出樣例：

```
{
  "ocr_text": "NIKE",
  "ocr_confidence": 0.92,
  "ocr_lines": [
    {"text": "NIKE", "score": 0.92, "polygon": [[10,10],[80,10],[80,30],
[10,30]]}
  ]
}
```

6.5 CLIP Logo Gate

Prompt：

- `a photo with a brand logo`

輸出樣例（quality_gate_json 內）：

```
{
  "clip_logo": {
    "score": 0.27,
    "threshold": 0.22,
    "soft_floor": 0.08,
    "decision": "passed"
  }
}
```

6.6 CLIP Brand Retrieval

輸出樣例：

```
{
  "clip_score": 0.86,
  "top_matches": [
    {"brand": "nike", "score": 0.86},
    {"brand": "adidas", "score": 0.63},
    {"brand": "new balance", "score": 0.57}
  ]
}
```

6.7 BLIP Caption

Prompt：

- `Describe the brand, object, and photo scene.`
- `Describe the logo, brand, and object in this crop.`

輸出樣例：

```
{
  "full_image_caption": "A person holding a red Coca-Cola can.",
  "logo_crop_caption": "The Coca-Cola logo on a soda can."
}
```

6.8 VLM (LLaVA)

Prompt：

- `Describe the brand, object, and photo scene.`
- `Describe the logo, brand, and object in this crop.`

輸出樣例：

```
{
  "full_image_caption": "A Nike sneaker displayed on a table.",
  "logo_crop_caption": "A white Nike swoosh on the shoe."
}
```

7. Quality Filter 決策規則（更細）

每張圖依序過五關，任一關失敗即標記 `quality_status=filtered`。

- 若來源可信且 CLIP score ≥ soft floor，會進入 soft pass（仍保留失敗原因）。
- pHash 去重優先於昂貴模型，避免重複算力浪費。
- 寫回欄位：`quality_status`, `quality_score`, `quality_gate_json`

8. SAM3 Two-Stage 流程

預設 object-first：

1. GroundingDINO object prompts → 物件 bbox
2. 對每個物件 crop → GroundingDINO logo prompts
3. 使用 logo bbox + SAM3 → mask

輸出欄位（segment records）：

- `object_grounding_bbox_xyxy`
- `object_grounding_label`
- `logo_grounding_bbox_xyxy`
- `logo_mask_path`

9. OCR / VLM / BLIP / Qwen 輸出位置

所有 OCR、Caption、Qwen 推理與 Attribution 結果都會寫入：`DB.logo_instances`，並輸出到 `export/knowledge_base.json`。

主要欄位：

- `ocr_text`, `ocr_confidence`
- `caption_text`, `caption_model`
- `attribution_json`（包含 clip_retrieval / captioning / qwen_knowledge / merged）
- `knowledge_json`（Qwen 多段式知識推理）
- `risk_json`（Qwen 風險判斷）

完整原始 payload：`logo_instances.provenance_json`

注意：`clip_retrieval` 現在預設只保留為診斷訊號， 不再直接主導最終品牌合併；最終品牌主要由 trusted metadata、OCR、caption / VLM 與 fallback similarity 決定。

9.1 Qwen 多段式推理

Qwen 不再一次問完整大題，而是拆成五段各自輸出 JSON：

1. `grounding`
2. `recognition`
3. `world_knowledge`
4. `risk`
5. `ambiguity`

好處是某一段失敗時，不會讓整個 knowledge/risk 全部變成 null。

最終寫回：

- `logo_instances.knowledge_json`
- `logo_instances.risk_json`
- `logo_instances.attribution_json.qwen_validation`

10. 輸出資料夾結構

```
results/<run_name>/
  meta/              # preflight/raw/candidate/passed
  fetch/             # records.json + images
  brand/             # brand_records.json
  segment/           # SAM3 outputs
  review/            # review_queue.json
  export/            # knowledge_base.json
  analysis/          # analysis.md + summary.json
  engine/            # main sqlite db
  staging/           # batch staging sqlite db
```

11. Knowledge Base 內容與結構（更詳細）

輸出檔：`export/knowledge_base.json`

結構為「每張影像一筆」，內含 logo_instances 與 brand_record：

```
{
  "image_id": "...",
  "engine_quality_status": "passed",
  "engine_quality_gate": {...},
  "engine_brand_record": {...},
  "logo_instances": [
    {"instance_id": "...", "bbox": [...], "attribution": {...},
"brand_record": {...}}
  ]
}
```

內容結構：

- **ImageRecord**：影像與來源資訊
- **LogoInstance**：bbox / mask / OCR / attribution 結果
- **BrandRecord**：品牌知識（industry / country / aliases）

核心欄位：

Entity	核心欄位
ImageRecord	image_id, image_url, local_image_path, source, source_channel, quality_status
LogoInstance	bbox_json, mask_path, ocr_text, clip_score, caption_text, attribution_json, knowledge_json, risk_json, tier
BrandRecord	brand_id, display_name, industry, country, aliases_en, tier

12. Schema 詳表

12.1 ImageRecord

欄位	型別	說明
image_id	TEXT	影像主鍵
external_key	TEXT	來源唯一 key
brand_hint	TEXT	品牌提示
category	TEXT	品類
source_name	TEXT	來源名稱
source_channel	TEXT	來源渠道
source_kind	TEXT	來源型態
scene_bucket	TEXT	場景桶
image_url	TEXT	原始 URL
local_image_path	TEXT	本地路徑
image_phash	TEXT	pHash
quality_status	TEXT	passed/filtered/pending
quality_score	REAL	綜合分數

欄位	型別	說明
quality_gate_json	TEXT(JSON)	各 gate 詳細結果
difficulty_flags_json	TEXT(JSON)	難例標籤
tier	TEXT	proposal/silver/gold

12.2 LogoInstance

欄位	型別	說明
instance_id	TEXT	logo instance 主鍵
image_id	TEXT	對應 image_id
brand_id	TEXT	推斷品牌
bbox_json	TEXT(JSON)	logo bbox
mask_path	TEXT	SAM3 mask 路徑
ocr_text	TEXT	OCR 文本
clip_score	REAL	CLIP brand 相似度
caption_text	TEXT	caption
caption_model	TEXT	caption model id
attribution_json	TEXT(JSON)	clip/caption/qwen/merged 詳細
knowledge_json	TEXT(JSON)	Qwen grounding/recognition/world knowledge/ambiguity
risk_json	TEXT(JSON)	Qwen risk reasoning
confidence	REAL	最終信心
review_status	TEXT	auto_accept/needs_review
tier	TEXT	proposal/silver/gold

12.3 BrandRecord

欄位	型別	說明
brand_id	TEXT	品牌主鍵
display_name	TEXT	品牌顯示名稱
industry	TEXT	產業
country	TEXT	國家
parent_company	TEXT	母公司
knowledge_json	TEXT(JSON)	品牌知識資料
tier	TEXT	silver/gold

13. JSON 實例樣本

13.1 raw_records.json（節錄）

```
[
  {
    "record_id": "nike_shoes_0001",
    "brand": "Nike",
    "category": "shoes",
    "image_url": "https://...",
    "source_name": "ad_site",
    "source_channel": "advertising",
    "source_kind": "marketplace"
  }
]
```

13.2 segment/records_with_logo_masks.json（節錄）

```
[
  {
    "record_id": "nike_shoes_0001",
```

```
    "logo_segmentation_status": "ok",
    "object_grounding_bbox_xyxy": [18,44,503,364],
    "logo_grounding_bbox_xyxy": [120,78,182,125],
    "logo_mask_path": ".../segment/masks/nike_shoes_0001.png"
  }
]
```

13.3 review/review_queue.json（節錄）

```
[
  {
    "instance_id": "img_001__logo_1",
    "review_status": "needs_review",
    "tier": "proposal",
    "ocr_text": "NIKE",
    "clip_score": 0.86
  }
]
```

13.4 export/knowledge_base.json（節錄）

```
[
  {
    "image_id": "img_001",
    "engine_quality_status": "passed",
    "logo_instances": [
      {
        "instance_id": "img_001__logo_1",
        "bbox": [120,78,182,125],
        "ocr_text": "NIKE",
        "clip_score": 0.86,
        "caption_text": "A Nike sneaker on a table."
      }
    ]
  }
]
```

14. 完整 Schema 欄位對應說明

以下說明 DB 欄位與輸出 JSON 的對應關係，便於確認每個欄位在 export 中的位置。

14.1 image_records → export/knowledge_base.json

DB 欄位	輸出 JSON 欄位	說明
image_id	image_id	影像主鍵
tier	engine_tier	影像 tier
quality_status	engine_quality_status	品質狀態
quality_score	engine_quality_score	品質分數
quality_gate_json	engine_quality_gate	各 gate 詳細
difficulty_flags_json	difficulty_flags	難例標記
brand_hint	engine_brand_record	對應品牌 record
raw_json	原始 record 欄位	展開到最外層

14.4 全 JSON 欄位逐欄說明（含型別/必填/驗證）

以下列出 export/knowledge_base.json 中每筆影像 record 的所有欄位說明。

JSON 欄位	型別	必填	驗證邏輯	來源/說明
record_id	string	yes	非空、唯一	collector 產生的 record id
brand	string	yes	非空字串	品牌文字
category	string	yes	允許值集合	品類
image_url	string	yes	URL 格式	圖片 URL
source_name	string	yes	非空	來源名稱
source_channel	string	yes	允許值集合	來源渠道

JSON 欄位	型別	必填	驗證邏輯	來源/說明
source_kind	string	yes	允許值集合	來源型態
capture_context	object	no	JSON 物件	拍攝上下文（國家/年份等）
image_id	string	yes	非空、唯一	影像主鍵
engine_tier	string	yes	proposal/silver/gold	影像 tier
engine_quality_status	string	yes	passed/filtered/pending	品質狀態
engine_quality_score	number	no	0~1	品質分數
engine_quality_gate	object	no	各 gate 子欄位完整	各 gate 詳細 JSON
difficulty_flags	array	no	string[]	難例標記
engine_brand_record	object	no	brand_record schema	對應 brand record
logo_instances	array	yes	LogoInstance[]	所有 logo instance 列表

logo_instances[] 欄位完整說明（含型別/必填/驗證）：

欄位	型別	必填	驗證邏輯	說明
instance_id	string	yes	非空、唯一	logo instance 主鍵
brand_id	string	no	需存在於 brand_records	推斷品牌
merged_brand_name	string	no	非空	合併後品牌名稱
detector_name	string	no	非空	使用的 detector
bbox	array	yes	長度=4	logo bbox
mask_path	string	no	路徑格式	SAM3 mask 路徑
ocr_text	string	no	可為空	OCR 文本
ocr_confidence	number	no	0~1	OCR 信心
clip_score	number	no	0~1	CLIP 相似度

欄位	型別	必填	驗證邏輯	說明
caption_text	string	no	可為空	caption 結果
caption_model	string	no	非空	caption 模型
attribution	object	no	必含 source/merged	完整 attribution JSON
confidence	number	no	0~1	最終信心
review_status	string	no	auto_accept/needs_review	人工審核狀態
tier	string	yes	proposal/silver/gold	資料 tier
provenance	object	no	JSON object	詳細來源與 payload
brand_record	object	no	brand_record schema	對應品牌 record

14.5 欄位驗證實作規則（正規化/允許值/檢查邏輯）

14.5.1 正規化邏輯

- **brand_id**：使用 `slugify`（小寫 + 非英數替換成 `-`）
- **source_channel/source_kind/category**：`strip().lower()`
- **scene_bucket**：優先 `capture_context.scene_type`，否則 `source_channel`，再否則 `category`
- **image_id**：`stable_hash(external_key, brand, category)`
- **instance_id**：`stable_hash(image_id, proposal)`

14.5.2 允許值列表（目前 pipeline 使用）

欄位	允許值/建議值	說明
tier	proposal / silver / gold	固定三態
review_statu	auto_accept / needs_review	目前 pipeline 使用

欄位	允許值/建議值	說明
s		
quality_status	passed / filtered / pending / error	quality gate 輸出
category	shoes / apparel / mugs / cans / smartphones	內建 prompts 支援
source_kind	official / marketplace / other	trusted catalog 判斷
source_channel	ecommerce / marketplace / advertising / editorial / social / sports_broadcast / user_capture	非硬性限制，主要用於 scene_bucket

14.5.3 Gate 驗證邏輯（實作細節）

- **尺寸**：width >= 256、height >= 256
- **格式**：JPEG/PNG/WEBP/JPG
- **模糊**：Laplacian variance >= 45.0
- **pHash**：distance <= 6
- **CLIP logo score**：score >= 0.22；trusted catalog 可 soft pass (0.08)
- **YOLO prescreen**：至少一個 bbox conf >= 0.3

14.5.4 BBox / Mask 基本驗證

- **bbox**：長度=4 且 x0 < x1, y0 < y1
- **mask_area_fraction**：0 ~ 1

14.5.5 正規化範例與合法/非法例子

欄位	正規化規則	合法例子	非法例子
brand_id	slugify（小寫+非英數轉 -）	输入: "New Balance" → `new-balance`	空字串 → `unknown`
source_channel	strip + lower	"Advertising " → `advertising`	null/空字串

欄位	正規化規則	合法例子	非法例子
source_kind	strip + lower	"Marketplace" → `marketplace`	非允許值（如 "foo"）
category	strip + lower	"Shoes" → `shoes`	不在清單（如 "car"）
scene_bucket	scene_type > source_channel > category	scene_type="stadium" → `stadium`	全部缺失 → `unknown`
image_id	stable_hash(external_key, brand, category)	固定輸入 → 固定輸出	brand/category 變動 → 新 id
bbox	[x0,y0,x1,y1] 且 x0<x1,y0<y1	[10,20,80,60]	[80,20,10,60]
mask_area_fraction	0 ~ 1	0.0032	-0.2 或 1.5

14.5.6 錯誤訊息範本與補救策略

欄位	錯誤訊息範本	補救策略
brand_id	`brand_id_invalid: empty or unknown`	回填 `brand` 原字串再做 slugify；若仍空則設 `unknown`
category	`category_invalid: not in allowed list`	映射到最近類別（如 "sneakers"→"shoes"），或保留原值並標記 difficulty_flags
source_channel	`source_channel_missing`	由 source_kind 或 source_name 推測，或設為 "unknown"
image_url	`image_url_invalid: not a URL`	使用 local_image_path，或標記 `missing_image`
bbox	`bbox_invalid: x0>=x1 or y0>=y1`	回退到 GroundingDINO 次佳候選框，或標記 `needs_review`
mask_area_fraction	`mask_fraction_invalid: out of range`	視為 mask 不可信，保留 bbox 但標記 `logo_mask_review_status=poor`

欄位	錯誤訊息範本	補救策略
quality_status	`quality_gate_failed: {reason}`	不進入 annotate；保留於 filtered 以供重跑調整閾值

14.5.7 驗證實作規則（對應程式邏輯）

- **slugify**：`re.sub(r"[^A-Za-z0-9._-]+","-", value.lower())`
- **brand_id**：`canonical_brand_id = slugify(name)`
- **image_external_key**：使用 record_id/brand/category/product_url/image_url 等組合做 `stable_hash`
- **image_id**：`stable_hash({external_key, brand, category})`
- **instance_id**：`stable_hash({image_id, proposal})`
- **scene_bucket**：`scene_type > source_channel > category`
- **trusted catalog**：`source_kind in {official, marketplace}` 或 `source_channel in {ecommerce, marketplace}`
- **tier**：允許值 `proposal/silver/gold`（TIER_VALUES）

14.2 logo_instances → export/knowledge_base.json.logo_instances

DB 欄位	輸出 JSON 欄位	說明
instance_id	logo_instances[].instance_id	logo instance 主鍵
brand_id	logo_instances[].brand_id	品牌 id
bbox_json	logo_instances[].bbox	bbox
polygon_json	logo_instances[].polygon	多邊形（可為 null）
rotated_box_json	logo_instances[].rotated_box	旋轉框（可為 null）
mask_path	logo_instances[].mask_path	SAM3 mask 路徑
ocr_text	logo_instances[].ocr_text	OCR 文本
clip_score	logo_instances[].clip_score	CLIP 相似度

DB 欄位	輸出 JSON 欄位	說明
caption_text	logo_instances[].caption_text	caption
attribution_json	logo_instances[].attribution	完整 attribution
knowledge_json	logo_instances[].knowledge	Qwen knowledge split
risk_json	logo_instances[].risk	Qwen risk split
provenance_json	logo_instances[].provenance	來源與詳細 payload

14.3 brand_records → export/knowledge_base.json.brand_record

DB 欄位	輸出 JSON 欄位	說明
brand_id	brand_record.brand_id	品牌主鍵
display_name	brand_record.display_name	顯示名稱
industry	brand_record.industry	產業
country	brand_record.country	國家
knowledge_json	brand_record.*	合併進輸出

15. 每個輸出 JSON 完整樣本

15.1 meta/raw_records.json

```
[
  {
    "record_id": "nike_shoes_0001",
    "brand": "Nike",
    "category": "shoes",
    "image_url": "https://example.com/img.jpg",
    "source_name": "brand_site",
    "source_channel": "advertising",
    "source_kind": "marketplace",
    "capture_context": {"country": "US"},
```

```
    "year": 2023
  }
]
```

15.2 meta/candidate_records.json

```
[
  {
    "record_id": "nike_shoes_0001",
    "brand": "Nike",
    "category": "shoes",
    "image_url": "https://example.com/img.jpg",
    "source_name": "brand_site",
    "source_channel": "advertising",
    "source_kind": "marketplace",
    "capture_context": {"country": "US"},
    "year": 2023
  }
]
```

15.3 meta/passed_records.json

```
[
  {
    "record_id": "nike_shoes_0001",
    "brand": "Nike",
    "category": "shoes",
    "image_url": "https://example.com/img.jpg",
    "quality_status": "passed",
    "quality_gate": {
      "size_format": "passed",
      "blur": "passed",
      "dedupe": "passed",
      "clip_logo": "passed",
      "logo_prescreen": "passed"
    }
  }
]
```

15.4 fetch/records.json

```
[
  {
    "record_id": "nike_shoes_0001",
    "brand": "Nike",
    "category": "shoes",
    "image_url": "https://example.com/img.jpg"
  }
]
```

15.5 brand/brand_records.json

```
[
  {
    "query": "Nike",
    "label": "Nike",
    "aliases_en": ["Just Do It"],
    "countries": ["United States"],
    "industries": ["Sportswear"],
    "matched": true
  }
]
```

15.6 segment/records_with_logo_masks.json

```
[
  {
    "record_id": "nike_shoes_0001",
    "logo_segmentation_status": "ok",
    "object_grounding_bbox_xyxy": [18,44,503,364],
    "object_grounding_label": "sneaker",
    "logo_grounding_bbox_xyxy": [120,78,182,125],
    "logo_mask_path": "segment/masks/nike_shoes_0001.png",
    "logo_mask_area_fraction": 0.0032
  }
]
```

15.7 review/review_queue.json

```
[
  {
    "instance_id": "img_001__logo_1",
    "review_status": "needs_review",
    "tier": "proposal",
    "ocr_text": "NIKE",
    "clip_score": 0.86
  }
]
```

15.7.1 Review UI 與決策狀態

Review UI 可直接顯示:

- 原圖
- bbox / mask overlay
- mask 原圖
- attribution / knowledge / risk / raw_json

Reviewer 可直接將 proposal 改成:

- `rejected`
- `silver`
- `gold`

實際落地欄位為: `logo_instances.review_status` 與 `logo_instances.tier`。

15.8 export/knowledge_base.json（更大規模示例）

```
[
  {
    "image_id": "img_001",
    "engine_quality_status": "passed",
    "engine_quality_gate": {"clip_logo": {"score": 0.27, "decision":
"passed"}},
    "logo_instances": [
```

```
      {
        "instance_id": "img_001__logo_1",
        "bbox": [120,78,182,125],
        "ocr_text": "NIKE",
        "clip_score": 0.86,
        "caption_text": "A Nike sneaker on a table.",
        "attribution": {"source": "ocr", "qwen_validation": {"valid": true,
"errors": []}},
        "knowledge": {
          "grounding": {"summary": "single visible logo on shoe side panel"},
          "recognition": {"brand": "Nike", "variant": "swoosh icon-only"},
          "world_knowledge": {"industry": "sportswear",
"scene_relation_type": "product_branding"},
          "ambiguity": {"needs_human_review": false, "uncertain_fields": []}
        },
        "risk": {
          "risk": {
            "primary_or_background": "primary",
            "commercial_risk": "third_party_brand_present",
            "recommended_action": "review_before_commercial_use"
          }
        }
      },
      {
        "instance_id": "img_001__logo_2",
        "bbox": [320,120,380,180],
        "ocr_text": "AIR",
        "clip_score": 0.61,
        "caption_text": "Secondary text mark on the shoe.",
        "attribution": {"source": "caption_hint"},
        "knowledge": {
          "grounding": {"summary": "secondary wordmark near heel"},
          "recognition": {"brand": "Nike", "sub_brand": "Air"},
          "world_knowledge": {"scene_relation_type": "product_branding"},
          "ambiguity": {"needs_human_review": true, "uncertain_fields":
["sub_brand"]}
        },
```

```
        "risk": {"risk": {"primary_or_background": "secondary",
"commercial_risk": "brand_exposure"}}
      }
    ]
  },
  {
    "image_id": "img_002",
    "engine_quality_status": "passed",
    "engine_quality_gate": {"clip_logo": {"score": 0.31, "decision":
"passed"}},
    "logo_instances": [
      {
        "instance_id": "img_002__logo_1",
        "bbox": [88,64,150,122],
        "ocr_text": "Coca-Cola",
        "clip_score": 0.92,
        "caption_text": "The Coca-Cola logo on a soda can.",
        "attribution": {"source": "clip+ocr"}
      }
    ]
  }
]
```


15.9 analysis/summary.json


```
{
  "fetched_records": 10,
  "knowledge_base_records": 10,
  "unique_brands": 7,
  "quality_passed": 10,
  "logo_instances": 8
}
```


15A. JSON 血緣與檔案歸屬

下表總結每個主要 JSON 的來源、下游與角色:

Path	Producer	Consumed By	Role
meta/preflight.json	preflight	human/debug	模型可用性診斷
meta/raw_records.json	collector-fetch-products --all	balance, ontology	原始候選池
meta/candidate_records.json	balance-records	ontology, ingest, gate	平衡後候選池
meta/passed_records.json	export-image-records	balance-records	quality gate 後乾淨池
fetch/records.json	collector 或 batch final balance	segment, phase1	最終主流程輸入集
brand/brand_records.json	ontology-fetch-brands	seed-ontology, phase1	品牌 ontology
segment/records_with_logo_masks.json	segment-sam3	phase1 --segment-records	bbox/mask 主 JSON
review/review_queue.json	review-queue	reviewer, review-apply	人工審核 queue
export/knowledge_base.json	export-kb	training/export	最終交付物
analysis/summary.json	analyze-run	dashboard/human	run 統計摘要

建議保留：`engine/logo_engine.db`, `fetch/records.json`, `brand/brand_records.json`, `review/review_queue.json`, `segment/records_with_logo_masks.json`, `export/knowledge_base.json`, `analysis/summary.json`。

16. 爬蟲方案與流程

16.1 Collector 策略來源

- **商品型來源**：品牌官網 / 商品頁（如 Adidas、Lululemon）
- **圖片型來源**：Unsplash、Flickr（API 友好）
- **資料集來源**：LAION / CSV index 篩選
- **外部野外圖**：手拍 / 社群 / sports / advertising（manifest 或 folder）

16.2 Collector 實際流程

1. 依照 brand × category 配對（或 manifest 清單）
2. 呼叫對應 collector（產品、外部資料、manifest）
3. 輸出 raw_records.json + 下載圖片
4. 進入 balance/ontology/ingest pipeline

16.3 Rate Limit 與重試

- 各來源具有不同限制，collector 內含 retry
- 若遇 403（如 Lululemon/Adidas），會 fallback 到下一策略

16.4 品牌/網頁候選名單位置與決策規則

候選名單定義於：

```
/raid/ming/logo/multi_brand_fetcher.py → BRAND_SOURCES
```

結構為：`(brand, category) → [strategy1, strategy2, ...]`

決策流程：

1. **--all**：直接取 `BRAND_SOURCES.keys()`
2. **--brands + --categories**：只保留存在於 `BRAND_SOURCES` 的 pair
3. 每個 pair 依策略順序嘗試；只要某策略回傳 ≥ min_records（預設 3）就停止

策略類型範例：

- `next_data`：抓 Next.js JSON
- `magento_bs4`：抓 Magento HTML
- `jsonld_per_page`：抓 JSON-LD
- `poshmark`：搜尋 fallback

17. 常見問題

17.1 SAM3 無法載入

確認:

```
/raid/ming/logo/logo_data_engine/sam3/sam3/model_builder.py
```

並確認 `meta/preflight.json` 內 SAM3 為 available。

17.2 VLM OOM

若 VLM 出現 CUDA OOM，請限制 GPU 或改用小模型。

10. 版本與維護

文件版本 v1.1 對應 pipeline 當前預設行為。